The documentation and process conversion measures necessary to comply with this revision shall be completed by 10 March 2019.

INCH-POUND

MIL-PRF-19500/683G
10 December 2018
SUPERSEDING
MIL-PRF-19500/683F
26 July 2016

PERFORMANCE SPECIFICATION SHEET

# TRANSISTOR, FIELD EFFECT, N-CHANNEL, RADIATION HARDENED, ENCAPSULATED (SURFACE MOUNT AND CARRIER BOARD PACKAGES), TYPE 2N7467, JANTXVR, F, G, AND H AND JANSR, F, G, AND H

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

## 1. SCOPE

1.1 Scope. This specification covers the performance requirements for an N-channel, enhancement-mode power MOSFET transistor with radiation hardened total dose and single event (SEE) effects ratings, with avalanche energy maximum rating ($E_{AS}$) and maximum avalanche current ($I_{AS}$). Two levels of product assurance (JANTXV and JANS) are provided for each encapsulated device. Provisions for radiation hardness assurance (RHA) to four radiation levels ("R", "F", "G", and "H") are provided for JANTXV and JANS product assurance levels. See 6.7 for JANHC and JANKC die versions.

1.2 Package outlines. The device package outlines are as follows: TO-276AC in accordance with figure 1, TO-276AC with lead option (U2L) in accordance with figure 2, SMD2 TO-276AC with carrier board option (U2S) in accordance with figure 3 for all encapsulated device types.

1.3 Maximum ratings. Unless otherwise specified, $T_A$ = +25°C.

| Type (1) | $P_T$ (2) $T_C$ = +25°C | $P_T$ $T_A$ = +25°C (free air) | $R_{\theta JC}$ (3) | $R_{\theta J}$ Carrier U2S | $R_{\theta J}$ Lid U2L (4) | $V_{DS}$ | $V_{DG}$ | $V_{GS}$ | $I_{D1}$ $T_C$ = +25°C (5) (6) | $I_{D2}$ $T_C$ = +100°C (5) (6) | $I_S$ | $I_{DM}$ (7) | $T_J$ and $T_{STG}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | W | W | °C/W | °C/W | °C/W | V dc | V dc | V dc | A dc | A dc | A dc | A(pk) | °C |
| 2N7467U2, U2L, U2S | 250 | 2.5 | 0.50 | 1.50 | 10 | 30 | 30 | ±20 | 75 | 75 | 75 | 300 | -55 to +150 |

(1) Electrical characteristics for the "U2L" and "U2S" suffix devices are identical to the non-suffix devices unless otherwise noted.
(2) Derate linearly 2.0 W/°C for $T_C$ > +25°C.
(3) See figure 4, thermal impedance curves.
(4) The Thermal resistance is applicable for mounting methods where a heatsink is attached to the lid for U2L suffix devices.
(5) The following formula derives the maximum theoretical $I_D$ limit. $I_D$ is limited by package and device construction to 75 Amps:

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

(6) See figure 5, maximum drain current graph.
(7) $I_{DM}$ = 4 X $I_{D1}$ as calculated in note (5).

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics at $T_C$ = +25°C.

| Type | Min $V_{(BR)DSS}$ $V_{GS}$ = 0 $I_D$ = 1.0 mA dc | $V_{GS(TH)}$ $V_{DS} \geq V_{GS}$ $I_D$ = 1.0 mA dc | | Max $I_{DSS1}$ $V_{GS}$ = 0 $V_{DS}$ = 80 percent of rated $V_{DS}$ | Max $r_{DS(ON)}$ (1) $V_{GS}$ = 12 V dc | | $E_{AS}$ at $I_{D1}$ | $I_{AS}$ |
|---|---|---|---|---|---|---|---|---|
| | | | | | $T_J$ = 25°C at $I_{D2}$ | $T_J$ = 150°C at $I_{D2}$ | | |
| | V dc | V dc | | µA dc | ohm | ohm | mJ | A |
| | | Min | Max | | | | | |
| 2N7467U2, U2L, U2S | 30 | 2.0 | 4.0 | 10 | .0035 | .006 | 500 | 75 |

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANTXV" and "JANS".

1.5.2 Radiation hardness assurance (RHA) designator. The RHA levels that are applicable for this specification sheet from lowest to highest are as follows: "R", “F”, “G”, and "H".

1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

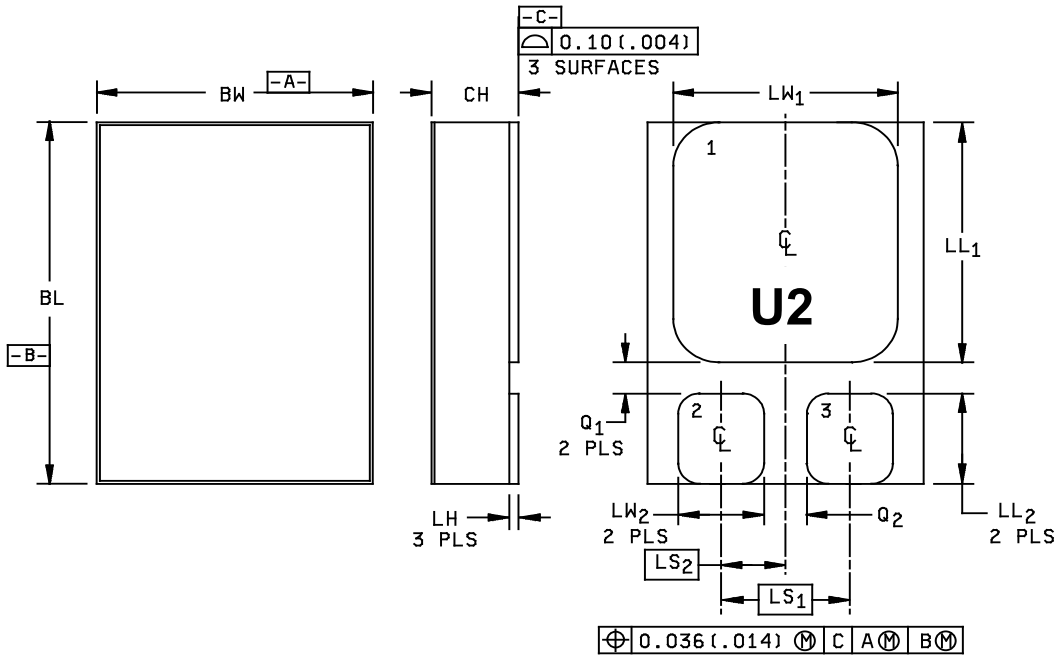
1.5.3.2 Second number symbols. The second number symbol for the transistors covered by this specification sheet is as follows: "7467".

1.5.3.3 Suffix letters. The suffix letters "U2" are used on devices that are packaged in the SMD2 TO-276AC package of figure 1. The suffix letters "U2L" are used on devices that are packaged in the SMD2 TO-276AC package and have additional flat leads added, see figure 2. The suffix letters "U2S" are used on devices that are packaged in the SMD2 TO-276AC package mounted to a carrier board, see figure 3.

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.



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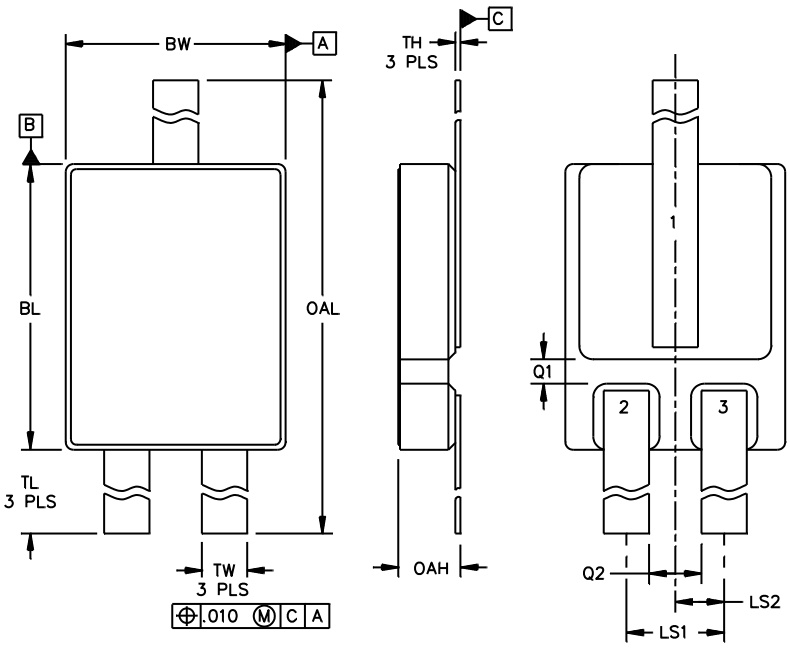

| LTR | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .685 | .695 | 17.40 | 17.65 |
| BW | .520 | .530 | 13.21 | 13.46 |
| CH | | .142 | | 3.61 |
| LL1 | .470 | .480 | 11.94 | 12.19 |
| LL2 | .152 | .162 | 3.86 | 4.11 |
| LH | .010 | .020 | .254 | .508 |
| LS1 | .240 BSC | | 6.10 BSC | |
| LS2 | .120 BSC | | 3.05 BSC | |
| LW1 | .435 | .445 | 11.05 | 11.30 |
| LW2 | .135 | .145 | 3.43 | 3.68 |
| Q1 | .035 | | .89 | |
| Q2 | .050 | | 1.27 | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Dimensions and tolerancing shall be in accordance with ASME Y14.5M.
4. Terminal 1 – Drain, Terminal 2 – Gate, Terminal 3 – Source

FIGURE 1. Physical dimensions for surface mount - 2N7467U2 (TO-276AC).



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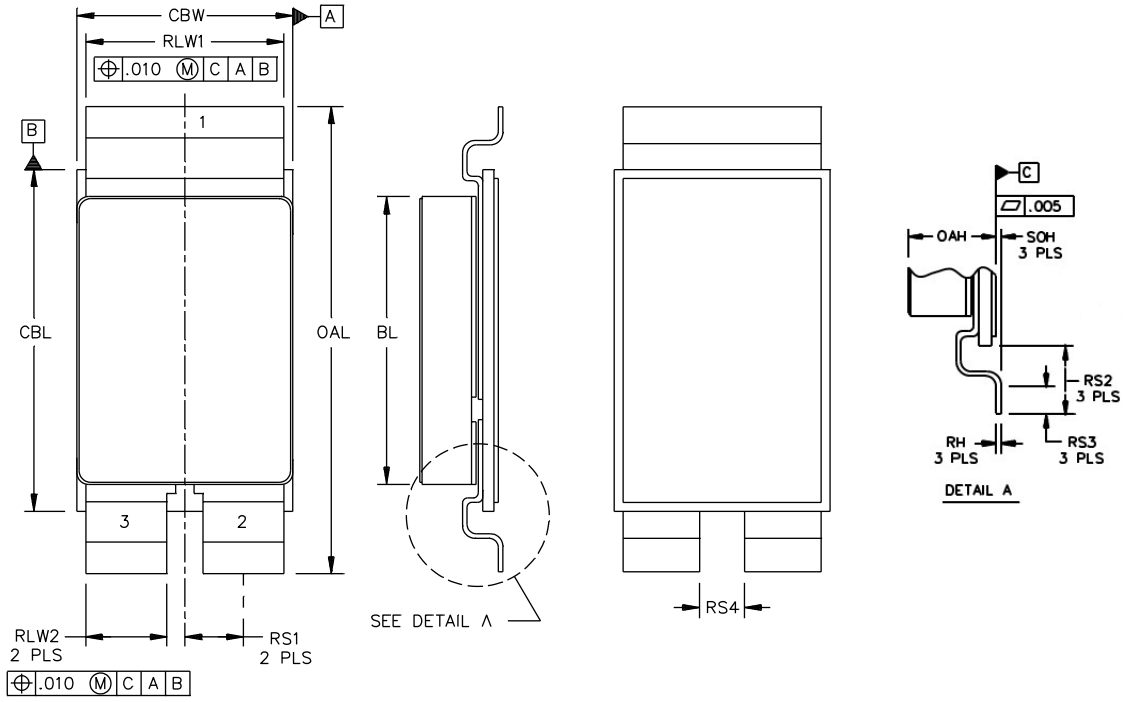

| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .685 | .695 | 17.40 | 17.65 |
| BW | .520 | .530 | 13.21 | 13.46 |
| LS1 | .240 BSC | | 6.10 BSC | |
| LS2 | .120 BSC | | 3.05 BSC | |
| Q1 | .035 | | 0.89 | |
| Q2 | .050 | | 1.27 | |
| TH | .005 | .007 | 0.127 | 0.177 |
| TL | .650 | .675 | 16.52 | 17.14 |
| TW | .095 | .105 | 2.42 | 2.66 |
| OAH | | .150 | | 3.81 |
| OAL | 1.985 | 2.045 | 50.42 | 51.94 |
| TERM 1 | Drain | | | |
| TERM 2 | Gate | | | |
| TERM 3 | Source | | | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The lid shall be electrically isolated from the drain, gate and source.
4. In accordance with ASME Y14.5M, diameters are equivalent to $\phi$x symbology.

FIGURE 2. Physical dimensions, U2 with leaded option (2N7467U2L).



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| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .685 | .695 | 17.40 | 17.65 |
| CBL | .825 | .840 | 20.96 | 21.34 |
| CBW | .520 | .535 | 13.21 | 13.59 |
| OAH | .174 | .204 | 4.42 | 5.18 |
| OAL | 1.109 | 1.144 | 28.17 | 29.06 |
| RH | .009 | .015 | 0.23 | 0.38 |
| RLW1 | .473 | .497 | 12.01 | 12.62 |
| RLW2 | .178 | .202 | 4.52 | 5.13 |
| RS1 | .1475 BSC | | 3.75 BSC | |
| RS2 | .142 | .152 | 3.61 | 3.86 |
| RS3 | .045 | .055 | 1.14 | 1.40 |
| RS4 | .093 | | 2.36 | |
| * SOH | .005 | .015 | 0.13 | 0.38 |
| TERM 1 | Drain | | | |
| TERM 2 | Gate | | | |
| TERM 3 | Source | | | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The lid shall be electrically isolated from the drain, gate and source.
4. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 3. Physical dimensions, U2 with carrier board option (2N7467U2S).



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## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

* DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* MIL-STD-883 - Test Method Standard Microcircuits

* (Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

$I_{AS}$..... Rated avalanche current, non-repetitive.
nC ..... nano Coulomb.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1 (U2, TO-276AC), 2 (U2L, surface mount TO-276AC with additional flat leads added) and 3 (U2S, surface mount TO-276AC with additional flat leads added and mounted to a carrier board) herein. Methods used for the electrical isolation of the terminals shall employ materials that contain a minimum of 90 percent $Al_2O_3$ (ceramic).

3.4.1 Lead material and finish. Terminal material shall be copper tungsten. Terminal finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of terminal finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction is not permitted to meet the requirements of this specification.



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3.4.3 Lead attach or Carrier package. Alternations to the device shall be performed on devices that have passed all screening and QCI required per MIL-PRF-19500 and listed herein. When leads or carrier attach is added to the U2 package, as a minimum, the vendor shall perform the tests specified in 4.3.4 herein.

* 3.4.4 Silicone die coat. The use of a silicone die coat requires a successful completion of MIL-STD-883, method 5011 on each epoxy lot for its intended applications, and as part of the full MIL-PRF-19500 qualification process.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking may be omitted from the body, but shall be retained on the initial container. Devices that have been altered with lead or carrier attached per the specification herein shall have the altered part number on the device or on the device packaging.

* 3.6 Electrostatic discharge sensitive (ESDS). The devices covered by this specification sheet have been classified as ESDS. The devices shall be handled in accordance with the ESD program established to comply with the requirements of MIL-PRF-19500 to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100\ k\Omega$, whenever bias voltage is to be applied drain to source.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.8 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.



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4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III (and table IV as applicable) herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.1.1 Single event effects (SEE). SEE shall be performed at initial qualification and after process or design changes which may affect radiation hardness (see table III and table V). Upon qualification, manufacturers shall provide the verification test conditions from section 5 of method 1080 of MIL-STD-750 that were used to qualify the device for inclusion into section 6 of the slash sheet. End-point measurements shall be in accordance with table II. SEE characterization data shall be made available upon request of the qualifying or acquiring activity.

4.2.1.2 Lead or carrier attach. For devices that include a lead or carrier attach package configuration qualification shall be performed in accordance with table IV herein, at initial qualification and after process or design changes.



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* 4.3 Screening (JANS, JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table E-IV of MIL-PRF-19500) (1) (2) | Measurement | |
|---|---|---|
| | JANS level | JANTXV level |
| (3) | Gate stress test (see 4.3.1) | Gate stress test (see 4.3.1) |
| (3) | Method 3470 of MIL-STD-750, $E_{AS}$ (see 4.3.2) | Method 3470 of MIL-STD-750, $E_{AS}$ (see 4.3.2) |
| (3) 3c | Method 3161 of MIL-STD-750, thermal impedance, (see 4.3.3) | Method 3161 of MIL-STD-750, thermal impedance, (see 4.3.3) |
| * 9 | Subgroup 2 of table I herein | Not applicable |
| 10 | Method 1042 of MIL-STD-750, test condition B | Method 1042 of MIL-STD-750, test condition B |
| * 11 | Subgroup 2 of table I herein;<br><br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater<br>$\Delta I_{DSS1}$ = ±5 μA dc or ±100 percent of initial value, whichever is greater | Subgroup 2 of table I herein; |
| 12 | Method 1042 of MIL-STD-750, test condition A | Method 1042 of MIL-STD-750, test condition A |
| 13 | Subgroups 2 and 3 of table I herein<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater<br>$\Delta I_{DSS1}$ = ±5 μA dc or ±100 percent of initial value, whichever is greater<br>$\Delta r_{DS(on)1}$ =±20 percent of initial value<br>$\Delta V_{GS(th)1}$ = ±20 percent of initial value | Subgroup 2 of table I herein<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater<br>$\Delta I_{DSS1}$ = ±5 μA dc or ±100 percent of initial value, whichever is greater<br>$\Delta r_{DS(on)1}$ = ±20 percent of initial value<br>$\Delta V_{GS(th)1}$ = ±20 percent of initial value |
| * 17 | Method 1081 of MIL-STD-750 (see 4.3.5). End-points: Subgroup 2 of table I herein. | Method 1081 of MIL-STD-750 (see 4.3.5). End-points: Subgroup 2 of table I herein. |

(1) At the end of the test program, $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ are measured).
(2) An out-of-family program to characterize $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, $V_{GS(th)1}$, and $r_{DS(ON)1}$ shall be invoked.
(3) Shall be performed anytime after temperature cycling, screen 3a; JANTXV level does not need to be repeated in screening requirements.



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4.3.1 Gate stress test. Apply $V_{GS}$ = 24 V minimum for t = 250 μs minimum.

4.3.2 Single pulse avalanche energy ($E_{AS}$).

a. Peak current ($I_{AS}$) ..... $I_{AS(max)}$.

b. Peak gate voltage ($V_{GS}$) ..... 12 V minimum up to rated $V_{GS}$ maximum.

c. Gate to source resistor ($R_{GS}$) ..... $25\Omega \leq R_{GS} \leq 200\Omega$.

d. Initial case temperature ($T_C$) ..... +25°C +10°C, -5°C.

e. Inductance (L)..... $\left[\frac{2E_{AS}}{(I_{D1})^2}\right]\left[\frac{V_{BR} - V_{DD}}{V_{BR}}\right]$ mH minimum.

f. Number of pulses to be applied ..... 1 pulse minimum.

g. Supply voltage ($V_{DD}$) ..... 25 V minimum up to rated $V_{GS}$ maximum.

* 4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{MD}$, $t_{SW}$, (and $V_H$ where appropriate). See table III, group E, subgroup 4 herein.

4.3.4 Lead or carrier attach screening (All quality levels). All surface mount devices with added leads or carrier boards shall be screened as specified herein.

| Screen | MIL-STD-750 Method | Conditions |
|---|---|---|
| 1. Hermetic Seal 1/<br>a. Fine Leak<br>b. Gross Leak | 1071 | |
| 2. Thermal Response (see 4.3.3)<br><br>A2 dc Electrical 2/ 3/ | 3161 | Read and Record. |
| 3. X-Radiography | 2076 | The solder material coverage at the package lead pad/SMD carrier sub interfaces shall be 85% minimum |
| 4. External Visual Examination | 2071 | Cracks or separation of materials shall not be evident on any device after the SMD lead attach assembly operation. Pad and Isolation areas shall be free from foreign matter and extraneous solder.<br>Solder filet coverage at the lead/package lead pad interfaces, along all visible sides, minimum of 75% solder fillet coverage. |
| 5a. Physical dimensions | 2066 | 6 piece sample, each device shall meet the requirements specified in figures 2 and 3. |
| 5b. Terminal Strength | 2036 | 3 piece sample. |

1/ Evaluation of surface sorption in accordance with method 1071 shall be performed.
2/ Only DC electrical test specified herein.
3/ When lead carrier bend is requested, the electrical test is performed prior to the bend process.



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* 4.3.5 Dielectric withstanding voltage.

a. Magnitude of test voltage ..... 900 V dc.

b. Duration of application of test voltage ..... 15 seconds (min).

c. Points of application of test voltage..... All leads to case (bunch connection).

d. Method of connection..... Mechanical.

e. Kilovolt-ampere rating of high voltage source ..... 1,200 V/1.0 mA (min).

f. Maximum leakage current..... 1.0 mA.

g. Voltage ramp up time..... 500 V/second.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500. End-point electrical measurements shall be in accordance with table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JANTXV) of MIL-PRF-19500, and as specified herein.

4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1051 | Test condition G, 100 cycles. |
| B3 | 2075 | See 3.4.2. |
| B3 | 2077 | Scanning electron microscope (SEM) qualification may be performed anytime prior to lot formation. |
| B4 | 1042 | Test condition D. Neither heat sink nor forced-air cooling on the device shall be permitted during the on cycle. The heating cycle shall be 60 seconds minimum. |
| B5 | 1042 | Test condition B, $V_{GS}$ = rated; $T_A$ = +175°C, t = 24 hours. |
| B5 | 1042 | Test condition A, $V_{DS}$ = rated; $T_A$ = +175°C; t = 120 hours. |
| B5 | 2037 | Bond strength; test condition D. |



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4.4.2.2 Quality level JANTXV, table E-VIB of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B2 | 1051 | Test condition G, 25 cycles. (45 total, including 20 cycles performed in screening). |
| B3 | 1042 | Test condition D. Neither heat sink nor forced-air cooling on the device shall be permitted during the on cycle. The heating cycle shall be 60 seconds minimum. |
| B4 | 2075 | See 3.4.2. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 1021 | Omit initial conditioning. |
| C2 | 2036 | Not applicable. |
| C5 | 3161 | Thermal resistance, see 4.3.3. |
| C6 | 1042 | Test condition D. Neither heat sink nor forced-air cooling on the device shall be permitted during the on cycle. The heating cycle shall be 60 seconds minimum. |

4.4.4 Group D Inspection. Group D inspection shall be conducted in accordance with appendix E, table E-VIII of MIL-PRF-19500 and table II herein.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III (and table IV as applicable) herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

* 4.5.2 Thermal resistance. The thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{MD}$, $t_{SW}$ (and $V_H$ where appropriate). See MIL-PRF-19500, table E-IX, group E, subgroup 4.



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* TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical inspection | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance 2/ | 3161 | See 4.3.3 | $Z_{\theta JC}$ | | | °C/W |
| Breakdown voltage, drain to source | 3407 | $V_{GS}$ = 0 V dc, $I_D$ = 1 mA dc, bias condition C | $V_{(BR)DSS}$ | 30 | | V dc |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1 mA dc | $V_{GS(TH)1}$ | 2.0 | 4.0 | V dc |
| Gate current | 3411 | $V_{GS}$ = +20 V dc bias condition C, $V_{DS}$ = 0 | $I_{GSSF1}$ | | +100 | nA dc |
| Gate current | 3411 | $V_{GS}$ = -20 V dc, bias condition C, $V_{DS}$ = 0 | $I_{GSSR1}$ | | -100 | nA dc |
| Drain current | 3413 | $V_{GS}$ = 0 V dc, bias condition C, $V_{DS}$ = 24 V dc | $I_{DSS1}$ | | 10 | µA dc |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = 12 V dc, condition A, pulsed (see 4.5.1), $I_D = I_{D2}$ | $r_{DS(on)1}$ | | .0035 | Ω |
| * Forward voltage | 4011 | Condition A, $I_D = I_{D1}$, $V_{GS}$ = 0 V dc | $V_{SD}$ | | 1.3 | V |
| Subgroup 3 | | | | | | |
| High-temperature operation: | | $T_C = T_J$ = +125°C | | | | |
| Gate current | 3411 | $V_{GS}$ = +20 V dc and -20 V dc, bias condition C, $V_{DS}$ = 0 | $I_{GSS2}$ | | ± 200 | nA dc |
| Drain current | 3413 | $V_{GS}$ = 0 V dc, bias condition C, $V_{DS}$ = 24 V dc | $I_{DSS2}$ | | 25 | µA dc |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = 12 V dc, condition A, pulsed (see 4.5.1), $I_D = I_{D2}$ | $r_{DS(on)3}$ | | .0055 | Ω |
| Gate to source voltage (thresholds) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1 mA dc | $V_{GS(TH)2}$ | 1.0 | | V dc |
| Low-temperature operation: | | $T_C = T_J$ = -55°C | | | | |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1 mA dc | $V_{GS(TH)3}$ | | 5.0 | V dc |

See footnotes at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 4 | | | | | | |
| Forward transconductance | 3475 | $I_D$ = 45 A, $V_{DD}$ = 15 V (see 4.5.1) | $g_{FS}$ | 45 | | |
| Switching time test | 3472 | $I_D$ = 45 A, $V_{GS}$= 12 V dc, $R_G$ = 2.35Ω, $V_{DD}$ = 15 V dc | | | | |
| Turn-on delay time | | | $t_{d(on)}$ | | 35 | ns |
| Rise time | | | $t_r$ | | 125 | ns |
| Turn-off delay time | | | $t_{d(off)}$ | | 80 | ns |
| Fall time | | | $t_f$ | | 50 | ns |
| Subgroup 5 | | | | | | |
| Safe operating area test (high voltage) | 3474 | See figure 6; $t_p$ = 10 ms $V_{DS}$ = 24 V | | | | |
| Electrical measurements | | See table I, subgroup 2 herein. | | | | |
| Subgroup 6 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 7 | | | | | | |
| Gate charge | 3471 | Condition B, $I_D$ = 45 A | | | | |
| On-state gate charge | | | $Q_{G(on)}$ | | 200 | nC |
| Gate to source charge | | | $Q_{GS}$ | | 55 | nC |
| Gate to drain charge | | | $Q_{GD}$ | | 40 | nC |
| * Reverse recovery time | 3473 | $d_i/d_t \leq$ 100 A/μs, condition A, $V_{DD} \leq$ 25 V, $I_D$ = 45 A | $t_{rr}$ | | 165 | ns |

1/ For sampling plan, see MIL-PRF-19500.

* 2/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 2 and 3 (JANTXV).
Group B, subgroups 3 and 4 (JANS).
Group C, subgroup 2 and 6.
Group E, subgroup 1.



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* TABLE II. Group D inspection.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Pre-irradiation limits | | Post-irradiation limits | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|
| | Method | Conditions | | R, F, G and H | | R, F and G | | H 4/ | | |
| | | | | Min | Max | Min | Max | Min | Max | |
| Subgroup 1 | | | | | | | | | | |
| Not applicable | | | | | | | | | | |
| Subgroup 2 | | $T_C$ = +25°C | | | | | | | | |
| Steady-state total dose irradiation ($V_{GS}$ bias) 5/ | 1019 | $V_{GS}$ = 12V<br>$V_{DS}$ = 0 | | | | | | | | |
| Steady-state total dose irradiation ($V_{DS}$ bias) 5/ | 1019 | $V_{GS}$ = 0<br>$V_{DS}$ = 80 percent of rated $V_{DS}$ (pre-irradiation) | | | | | | | | |
| End-point electricals: | | | | | | | | | | |
| Breakdown voltage, drain to source | 3407 | Bias condition C<br>$V_{GS}$ = 0<br>$I_D$ = 1 mA | $V_{(BR)DSS}$ | 30 | | 30 | | 30 | | V dc |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$ | $V_{GSth1}$ | 2.0 | 4.0 | 2.0 | 4.0 | 1.5 | 4.0 | V dc |
| Gate current | 3411 | Bias condition C<br>$V_{GS}$ = 20 V<br>$V_{DS}$ = 0 | $I_{GSSF1}$ | | 100 | | 100 | | 100 | nA dc |
| Gate current | 3411 | Bias condition C<br>$V_{GS}$ = -20 V<br>$V_{DS}$ = 0 | $I_{GSSR1}$ | | -100 | | -100 | | -100 | nA dc |
| Drain current | 3413 | Bias condition C<br>$V_{GS}$ = 0<br>$V_{DS}$ = 80 percent of rated $V_{DS}$ (pre-irradiation) | $I_{DSS1}$ | | 10 | | 10 | | 25 | µA dc |

See footnotes at end of table.



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* TABLE II. Group D inspection - Continued.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Pre-irradiation limits | | Post-irradiation limits | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|
| | Method | Conditions | | R, F, G and H | | R, F and G | | H 4/ | | |
| | | | | Min | Max | Min | Max | Min | Max | |
| Subgroup 2 -Continued | | | | | | | | | | |
| Static drain to source on-state voltage 6/ | 3405 | Condition A $V_{GS}$ = 12 V pulsed (see 4.5.1) $I_D$= 45 A | $V_{Dson1}$ | | .180 | | .180 | | .2025 | V dc |
| Static drain to source on-state voltage | 3405 | Condition A $V_{GS}$ = 12 V pulsed (see 4.5.1) $I_D$= 45 A | $V_{Dson1}$ | | .1575 | | .1575 | | .180 | V dc |
| * Forward voltage source to drain diode | 4011 | Condition A, $V_{GS}$ = 0, $I_D$= 45 A | $V_{SD}$ | | 1.3 | | 1.3 | | 1.3 | V dc |

1/ For sampling plan, see MIL-PRF-19500.
2/ Group D qualification may be performed anytime prior to lot formation. Wafers qualified to these group D QCI requirements may be used for any other specification utilizing the same die design.
3/ At the manufacturer's option, group D samples need not be subjected to the screening tests, and may be assembled in its' qualified package or in any qualified package that the manufacturer has data to correlate the performance to the designated package.
4/ The "H" designation represents devices which pass end-points at the G, R, and F designated Total-Ionizing-Dose (TID).
5/ Separate samples shall be pulled for each bias.
6/ Limit using TO-204AE package. The higher package resistance necessitates the higher $V_{DSon1}$ limit when the manufacturer uses the alternate package as allowed in 4/ above.



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* TABLE III. Group E inspection (all quality levels) for qualification or re-qualification only.

| Inspection | MIL-STD-750 | | Qualification and large lot quality conformance inspection |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices c = 0 |
| Temperature cycling | 1051 | Test condition G, 500 cycles. | |
| Hermetic seal<br>Fine leak<br>Gross leak | 1071 | As applicable. | |
| Electrical measurements | | See table I, subgroup 2. | |
| Subgroup 2 1/ | | | 45 devices c = 0 |
| Steady-state gate bias | 1042 | Condition B, 1,000 hours. | |
| Electrical measurements | | See table I, subgroup 2. | |
| Steady-state reverse bias | 1042 | Condition A, 1,000 hours. | |
| Electrical measurements | | See table I, subgroup 2. | |
| Subgroup 4 | | | Sample size N/A |
| Thermal impedance curves | | See MIL-PRF-19500. | |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 10 | | | 22 devices c = 0 |
| Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors | 3476 | Test conditions shall be derived by the manufacturer | |
| Subgroup 11 | | | 3 devices |
| SEE 2/ 3/ | 1080 | See MIL-STD-750 method 1080 and 6.2. | |

1/ A separate sample for each test shall be pulled.
2/ Group E qualification of SEE testing may be performed prior to lot formation. Qualification may be extended to other specification sheets utilizing the same structurally identical die design.
3/ Device qualification to a higher level linear energy transfer (LET) is sufficient to qualify all lower level LETs.



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TABLE IV. Lead alternation Qualification inspection requirements.

| Inspections 1/ | MIL-STD-750 | | Sample size |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 6 devices, c = 0 |
| Temperature cycle | 1051 | 100 Temp cycles, test condition G or maximum storage temperature. | |
| Hermetic seal | 1071 | | |
| Fine leak<br>Gross leak | | | |
| A2 dc electrical | | Read and record. | |
| Thermal response | 3161 | | |
| External visual examination | 2071 | Cracks or separation of materials shall not be evident on test samples. | |
| Subgroup 2 | | | 6 devices, c = 0 |
| Intermittent operating life | 1042 | Condition D; 6,000 cycles. | |
| A2 dc electrical | | Read and record. | |
| Thermal response | 3161 | | |
| External visual examination | 2071 | Cracks or separation of materials shall not be evident on test samples. | |
| Subgroup 3 | | | 6 devices, c = 0 |
| Terminal strength | 2036 | Tension; Condition A 10lbs for 10 seconds<br>Fatigue; Condition E 3 arcs of 90 +/_5 degrees each 8.0 oz. | |
| A2 dc electrical | | Read and record. | |
| External visual examination | 2071 | Cracks or separation of materials shall not be evident on test samples. | |

1/ Qualification samples performed on non-formed leaded devices.



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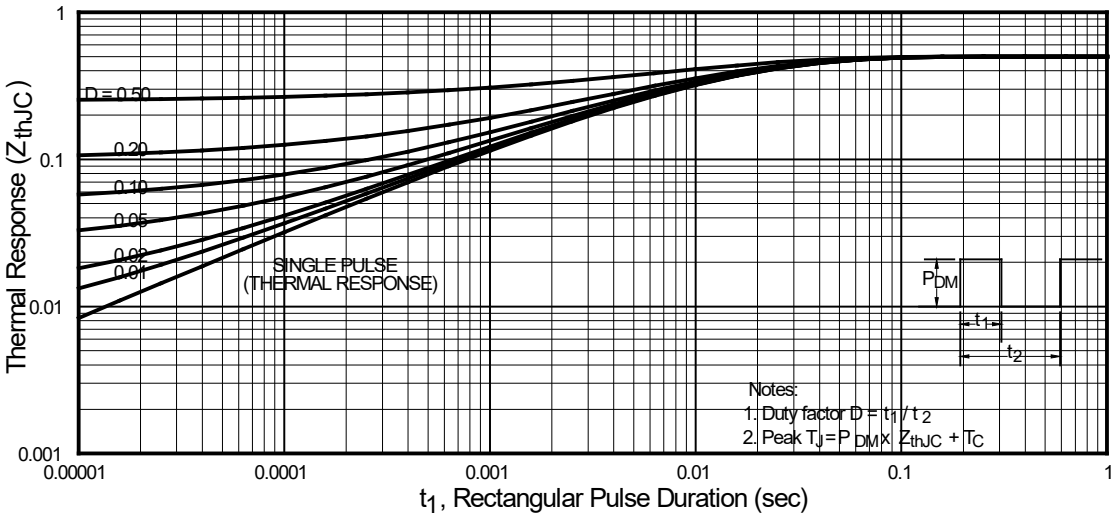



FIGURE 4. Thermal impedance curves.

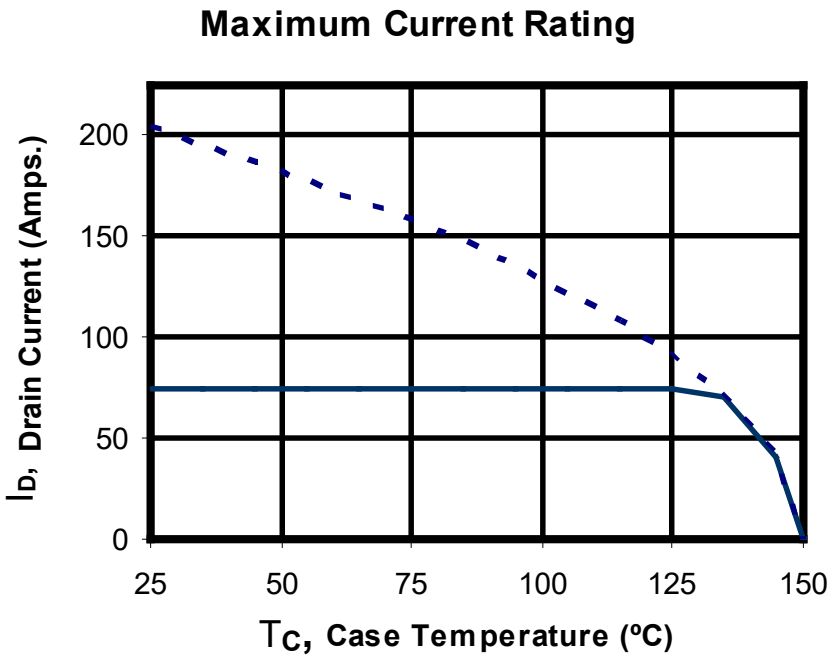



FIGURE 5. Maximum drain current versus case temperature graph.



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2N7467U2, U2L, U2S

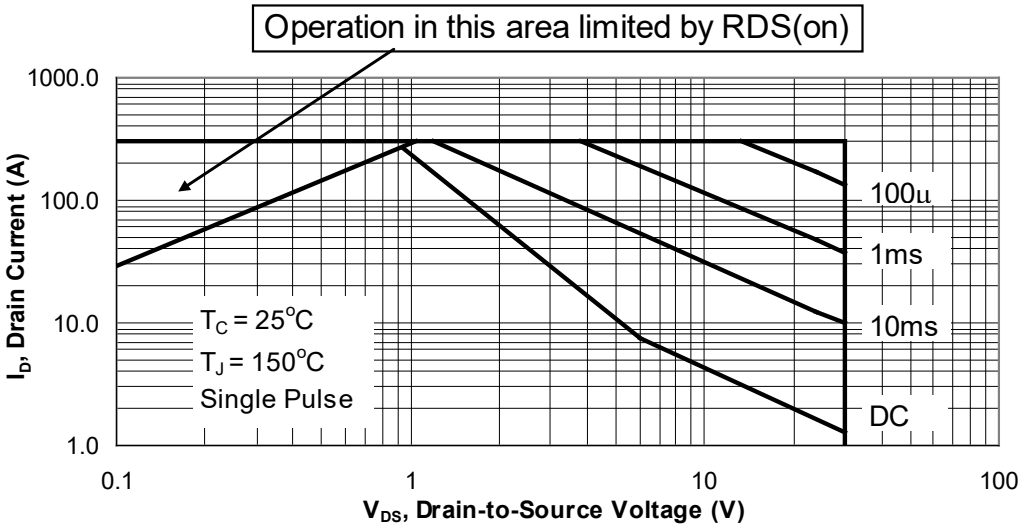



FIGURE 6. Safe operating area graph.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.5.

e. For acquisition of RHA designated devices, table II, subgroup 1 testing of group D herein is optional. If subgroup 1 is desired, it should be specified in the contract or order.

f. If SEE testing data is desired, it should be specified in the contract or order.

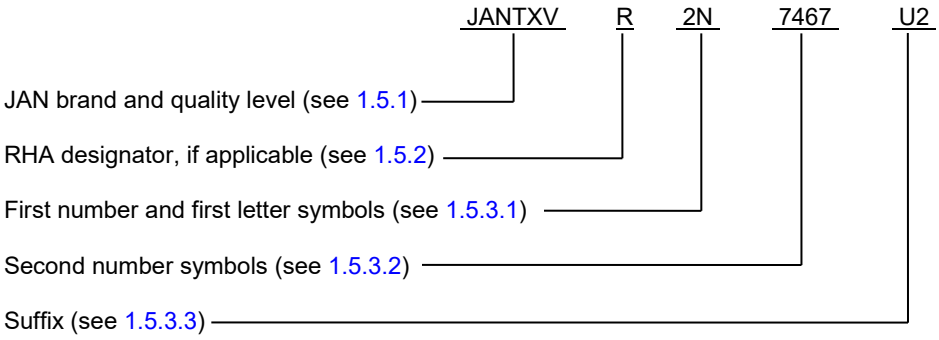
g. If specific SEE characterization conditions are desired (see section 6.8 and table V), manufacturer's cage code should be specified in the contract or order.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.



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6.4 PIN construction example. The PINs for encapsulated devices are construction using the following form.




6.5 List of PINs. The following is a list of possible PINs available on this specification sheet.

| PINs for devices of the "TXV" quality level | PINs for devices of the "TXV" quality level with RHA (1) | PINs for devices of the "S" quality level | PINs for devices of the "S" quality level with RHA (1) |
|---|---|---|---|
| JANTXV2N7467U2 | JANTXV#2N7467U2 | JANS2N7467U2 | JANS#2N7467U2 |
| JANTXV2N7467U2L | JANTXV#2N7467U2L | JANS2N7467U2L | JANS#2N7467U2L |
| JANTXV2N7467U2S | JANTXV#2N7467U2S | JANS2N7467U2S | JANS#2N7467U2S |

(1) The number sign (#) represents one of four RHA designators available on this specification sheet ("R" "F", "G" or "H").

6.6 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's Part or Identifying Number (PIN). This information in no way implies that manufacturer's PIN's are suitable for the military PIN.

| Preferred types | Commercial types |
|---|---|
| 2N7467U2 | IRHNA57Z60 |

6.7 JANHC and JANKC die versions. The JANHC and JANKC die versions of these devices are covered under specification sheet MIL-PRF-19500/741.

6.8 Application data.

6.8.1 Manufacturer specific irradiation data. Each manufacturer qualified to this slash sheet has characterized its devices to the requirements of MIL-STD-750 method 1080 and as specified herein. Since each manufacturer's characterization conditions can be different and can vary by the version of method 1080 qualified to, the MIL-STD-750 method 1080 revision version date and conditions used by each manufacturer for characterization have been listed here (see table IV) for information only. SEE conditions and figures listed in section 6 are current as of the date of this specification sheet, please contact the manufacturer for the most recent conditions.



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TABLE V. Manufacturers characterization conditions.

| Manufactures cage | Inspection | MIL-STD-750 | | Sample plan |
|---|---|---|---|---|
| | | Method | Conditions | |
| 69210 (Applicable to devices with a date code of 21 August 2012 and older) | SEE 1/ | 1080 | See MIL-STD-750E method 1080.0 dated 20 November 2006.<br>See figure 7. | |
| | Electrical measurements | | $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ in accordance with table I, subgroup 2 | 3 devices |
| | SEE irradiation: | | Fluence = 3E5 ±20 percent ions/cm$^2$<br>Flux = 2E3 to 2E4 ions/cm$^2$/sec, temperature = 25º ±5 °C<br><br>Surface LET = 38 MeV-cm2/mg ±5%,<br>range = 38 µm ±7.5%,<br>energy = 300 MeV ±7.5%.<br>In situ bias conditions: $V_{DS}$ = 30 V and $V_{GS}$ = -10 V,<br>$V_{DS}$ = 22.5 V and $V_{GS}$ = -15 V,<br>$V_{DS}$ = 15 V and $V_{GS}$ = -20 V,<br>(nominal 3.86 MeV/nucleon at Brookhaven National Lab Accelerator).<br><br>Surface LET = 61 MeV-cm2/mg ±5%,<br>range = 31 µm ±10%,<br>energy = 330 MeV ±7.5%.<br>In situ bias conditions: $V_{DS}$ = 25 V and $V_{GS}$ = -5 V,<br>$V_{DS}$ = 20 V and $V_{GS}$ = -10 V,<br>$V_{DS}$ = 15 V and $V_{GS}$ = -15 V,<br>$V_{DS}$ = 7.5 V and $V_{GS}$ = -20 V,<br>(nominal 2.92 MeV/nucleon at Brookhaven National Lab Accelerator).<br><br>Surface LET = 84 MeV-cm2/mg ±5%,<br>range = 28 µm ±7.5%,<br>energy = 350 MeV ±7.5%.<br>In situ bias conditions: $V_{DS}$ = 25 V and $V_{GS}$ = -5 V,<br>$V_{DS}$ = 20 V and $V_{GS}$ = -10 V,<br>(nominal 1.98 MeV/nucleon at Brookhaven National Lab Accelerator). | |
| | Electrical measurements | | $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ in accordance with table I, subgroup 2 | |
| Upon qualification, all manufacturers should provide the verification test conditions to be added to this table. | | | | |

1/ $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ was examined before and following SEE irradiation to determine acceptability for each bias condition. Other test conditions in accordance with table I, subgroup 2, may be performed at the manufacturer's option.



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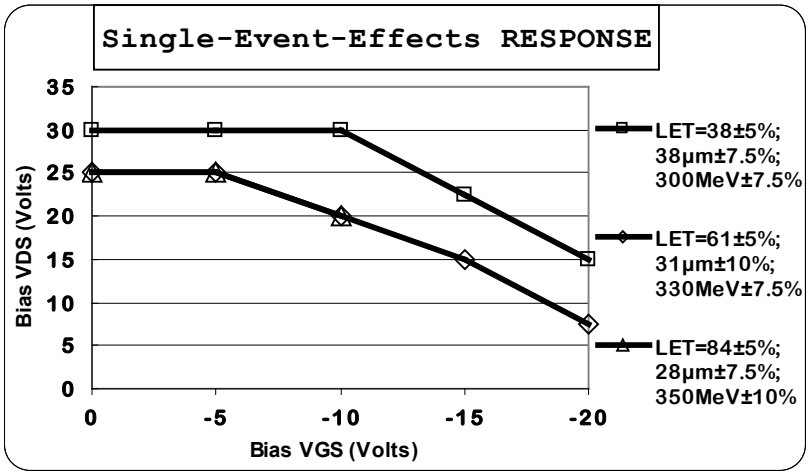



FIGURE 7. Cage 68210 typical SEE response graph.

6.9 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218–3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 693-1642 or DSN 850-6939.

6.10 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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